

2025Year 2nd Semester Syllabus

Created Date	2025-08-08 11:01:30		Last-Modified	2025-08-20 16:03:13
Course Title	REAL-TIME SYSTEMS		Course Code-Section	CAS4155-01
Credit/Time/ Experiment, Lab, Practical Technique Time	3/Thu2,3,Fri2		Department	School of Computing
Time	Thu2,3,Fri2		Location	EngHD508
Exam Date & Time	Midterm exam		Final exam	
Class Language	English		Evaluation Type	Absolute evaluation

Instructor's Profile	Name	Lee Jinkyu	Contact Information	Telephone	
	Department	DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING		Mail	JINKYU.LEE@YONSEI.AC.KR
	Office			Interview information	Fri11-12 & 13-14 (by appointment)

TA's Name & Contact Information	Name		Contact Information	Telephone	
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Course Description Brief Introduction of the Course	As many computing systems start to operate autonomously due to developments such as Internet-of-things (IoT) and Cyber-Physical Systems (CPS), the requirements for the application's completion time are important. For example, the life of the driver/pedestrian is at risk if an object detection task for an autonomous vehicle cannot be completed within a given deadline. In this way, a system that provides time predictability of the execution of a given task is called a real-time system, and is used in safety-critical and mission-critical industries such as autonomous vehicles, avionics, space, medical devices, robots and national defense. This course deals with the overall design and analysis of real-time systems, and specifically studies the concept of real-time systems, related problems, their approaches, and analysis methods. In addition, this course expands the systems/principles of computer science/engineering towards timing guarantees, so that we study the principles of programming for real-time systems, the design and functions of real-time operating systems, the operation of real-time networks, and emerging applications that require timing guarantees (e.g., machine learning related to autonomous driving).				
Course Goals	1.	Korean	실시간 시스템이 무엇인지 그리고 기존 컴퓨팅 시스템과의 차이 이해	20%	
		English	Understanding what real-time systems are and how they differ from conventional computing systems		
	2.	Korean	타이밍보장(실시간성 보장)을 위한 알고리즘과 스케줄링에 대해 이해	50%	
		English	Understanding algorithms and scheduling for guaranteeing timing constraints (real-time guarantees)		
	3.	Korean	타이밍 보장이 요구되는 다양한 응용 및 시스템의 활용	30%	
		English	Utilizing various applications/systems subject to timing guarantees		
	4.	Korean		0%	
		English			
	5.	Korean		0%	
		English			

Core Competencies	The total measurable competencies must be 100%. Each course objective should set the competency as 25%. The core and major competencies should equal at least 50%.									
Sub-Competencies/Learning Unit1										
Sub-Competencies/Learning Unit2										
Sub-Competencies/Learning Unit3										
Core Competencies(Liberal Arts)/Major competency(	Must reflect the interrelationship between core competencies (elective courses) and major competencies (major studies).									
Sustainable Development Goals										
Average Recommended Amount of Learning per	Average Reading Volume						Average amount of writing(Based on A4)			
Course Methods (%) Total Amount 100	Lecture		Practice Training		Presentation		Dabate		Team Project	
	80%		0%		20%		0%		0%	
Course Methods 2 Select Relevant Items	PBL Subject		Capstone Design		CBL, Social Innovation Course		Flipped Classroom		Work Experience, Internsh	
Grading Policy(%) Total Amount 100 Free Input for Other Information	Midterm exam	Final exam	Quiz	Individual Assignment	Team Assignment	Attendance	Others			
	20%	40%	0%	20%	0%	10%	10%			
Assignment/ Report, Project Guide	Title of Assignment/Project Name, and Method of Filling Out				Submission Deadline		Type of Submission and Method			
Prerequisite	Knowledge of Operating Systems and Algorithm				Online Course Address					

Course Material	Course Material Name	Author	Publisher	Publish Year	ISBN
참고자료	Real-Time Systems	Jane W. S. Liu	Prentice Hall	2000	
참고자료	Analysable Real-Time Systems: Programmed in Ada	A. Burns and A. Wellings	Addison Wesley	2009	
참고자료	Hard Real-Time Computing Systems	Giorgio C. Buttazzo	Springer	2011	

Main Learner Precautions	
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Attachment	
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Weekly Plan

week	Period	Weekly Topic & Contents	Remarks
1		Course introduction	(9.1.) Fall semester classes begin (9.3. - 9.5.) Course add and drop period
2		Basic concepts	
3		Sample applications	
4		Modeling time-critical tasks	
5		Scheduling time-critical tasks	(10.3.) National Foundation Day
6		Scheduling time-critical tasks	(10.5. - 10.7.) Chuseok Holidays (10.8.) First third of the semester ends (10.9.) Hangeul Proclamation Day
7		Reserved	(10.13. - 10.15.) Course withdrawal period
8		Mid-term Exam	(10.20. - 10.26.) Midterm Examinations
9		Scheduling periodic time-critical tasks	(10.27. - 10.29.) Application Period for S/U evaluation
10		Scheduling periodic time-critical tasks	
11		Scheduling periodic time-critical tasks	(11.13.) Second third of the semester ends
12		Resource sharing and task synchronization	
13		Real-time + machine learning	
14		Real-time + operating systems	
15		Reserved	(12.8. - 12.14.) Self-study
16		Final Exam	(12.15. - 12.21.) Final Examinations

계절제 수업계획서 작성 시 이전 계절제 수업계획서를 복사하시거나, 직접 입력하시기 바랍니다.

정규학기 수업계획서를 복사하는 경우 오류가 발생할 수 있습니다.

- Students with disabilities(SWDs) can request accommodations related to lectures, assignments, or tests by contacting the course professor at the beginning of semester.

(However, accommodations may vary depending on the essentiality of lecture and discretion of professors.)

[Lecture]

- Visual Impairment: alternative, braille, enlarged reading materials, note-taker
 - Physical Impairment: alternative reading materials, access to classroom, note-taker, assigned seat
 - Hearing Impairment: note-taker/stenographer, recording lecture
 - Intellectual Disability/Autism: note-taker
- [Assignments and Test]
- Visual/Physical/Hearing Impairment: (reasonable) extra days for submission, alternative type of assignment, extended test time, alternative type of test, arranging separate test room, and proctors, test ghostwriter
 - Intellectual Disability/Autism: (reasonable) extra days for submission, alternative type of assignment